

*Supporting Information for***Ultra-compact and ultra-high speed topological photonic crystal modulator based on thin film lithium niobate**

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1. ILs of TPhC modulators with different electrode gaps

For our device, there are two main reasons for the IL. Firstly, the process imperfection will result in surface roughness and thus will cause large propagation loss. Secondly, the cavity with different Q factor will also result in different IL. During the experiment, we have designed cavity with different topological photonic crystal unit cell number (N). We found that the Q factor of the cavity will increase when the cell number increase. And the IL will also increase correspondingly.

The measured insertion loss (IL) and tuning efficiency of the TPhC modulators with varying electrode gaps are presented in Figure S1. Within a specific range of electrode spacings, decreasing the gap leads to minimal changes in the modulator's insertion loss, while significantly enhancing its tuning efficiency.

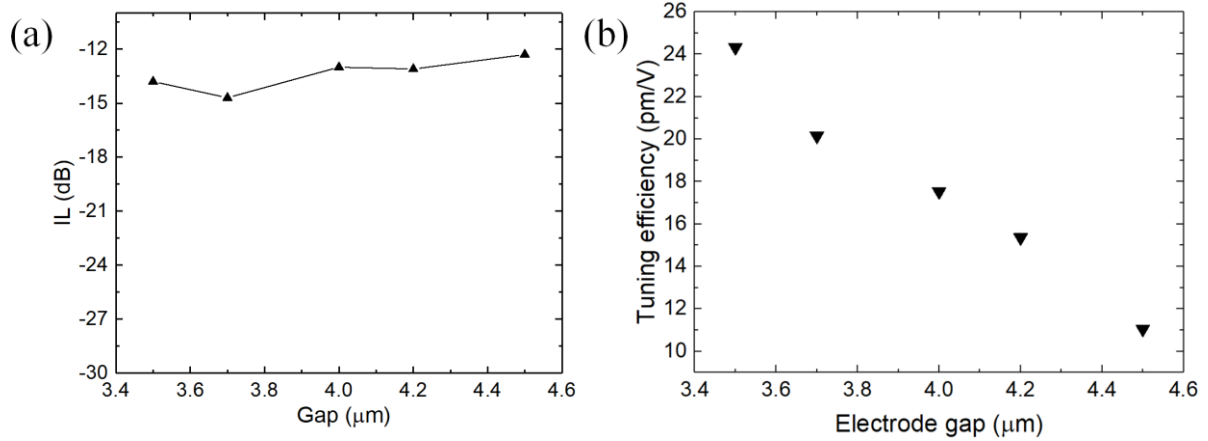


Figure S1. A comparison of the ILs under varied electrode separations. The measured (a) IL and (b) tuning efficiencies of TPhC modulators with different electrode gaps from 3.5 to 4.5 μm .

2. Tuning efficiencies of TPhC modulators with different electrode gaps

The measured electro-optic (EO) tuning efficiencies of TPhC modulators with electrode gaps ranging from 3.5 μm to 4.5 μm are presented in Figure S2. The results indicate that reducing the electrode spacing leads to a notable enhancement in tuning efficiency. This improvement arises from the intensified electric field confined within the waveguide region between the electrodes, which strengthens the overlap between the optical mode and the TFLN, thereby promoting more efficient EO interaction.

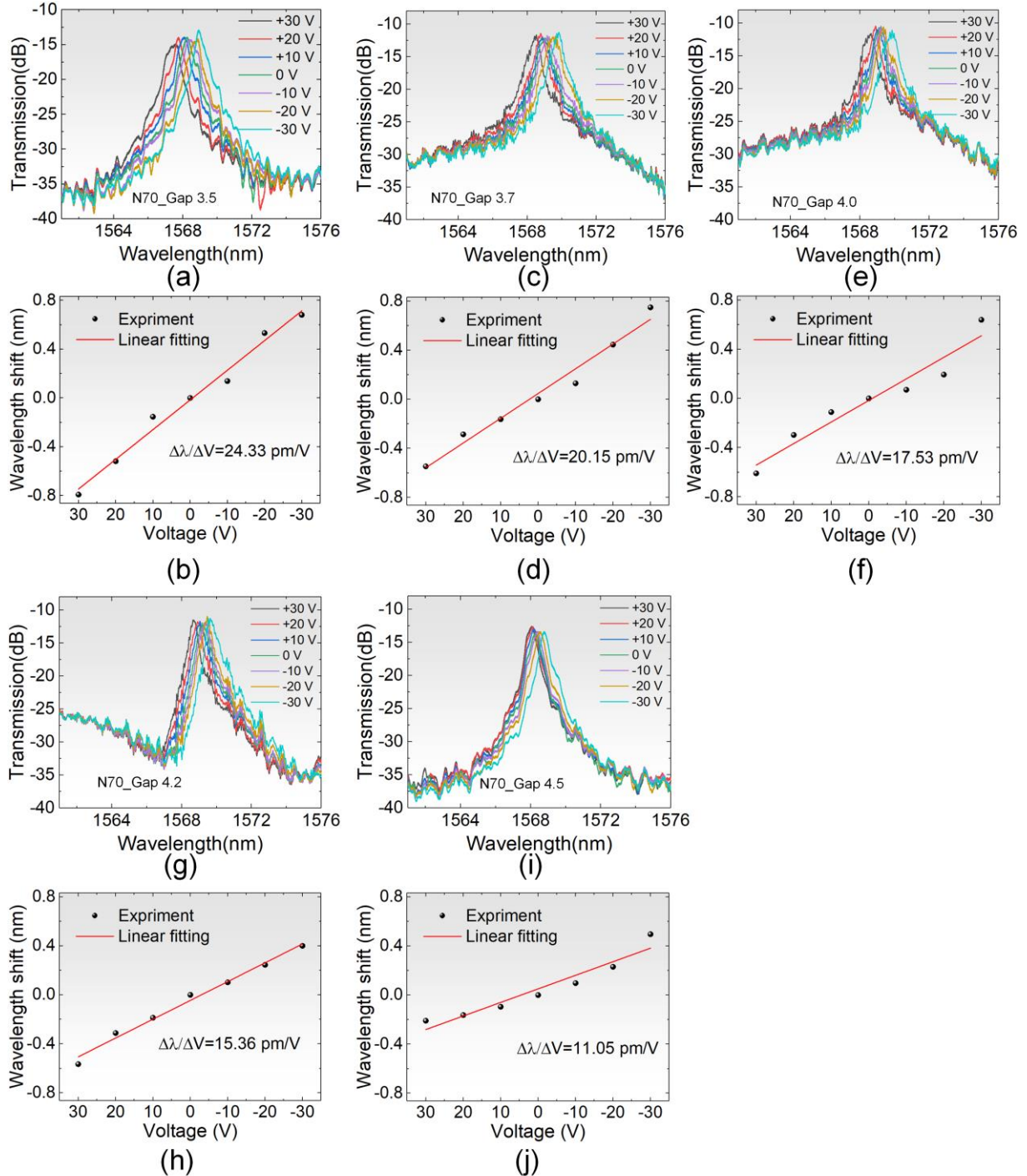


Figure S2. A comparison of the tuning efficiencies under varied electrode separations. The measured transmission spectra at different voltages and the corresponding tuning efficiencies of the TPhC modulator with 70 unit cells and varying electrode gaps are shown in pairs: (a) and (b) for a 3.5 μm gap, (c) and (d) for a 3.7 μm gap, (e) and (f) for a 4.0 μm gap, (g) and (h) for a 4.2 μm gap, and (i) and (j) for a 4.5 μm gap. In each pair, the first plot (a, c, e, g, i) represents the transmission spectra at different voltages, while the second plot (b, d, f, h, j) shows the tuning efficiency.

3. A comparative analysis of the process flows for directly etched and heterogeneously integrated schemes

Our proposed topological photonic crystal (TPhC) modulator were fabricated on an 400 nm X-cut lithium niobate on insulator (LNOI) wafer. The LN layer is etched using inductive couple plasma (ICP). And electrode is deposited using electro beam evaporation followed by a lift-off process, as shown in Figure S3. (a). The process flow of the heterogeneously integrated modulator is more complex, as shown in Figure S3. (b). Firstly, a layer of silicon nitride (SiN) thin film is deposited on an LNOI wafer. Subsequently, planarization processes and performance characterization are conducted to achieve a flat SiN film with desired thickness. Finally, the TPhC waveguide are etched on the SiN layer, followed by electrode deposition and lift-off process.

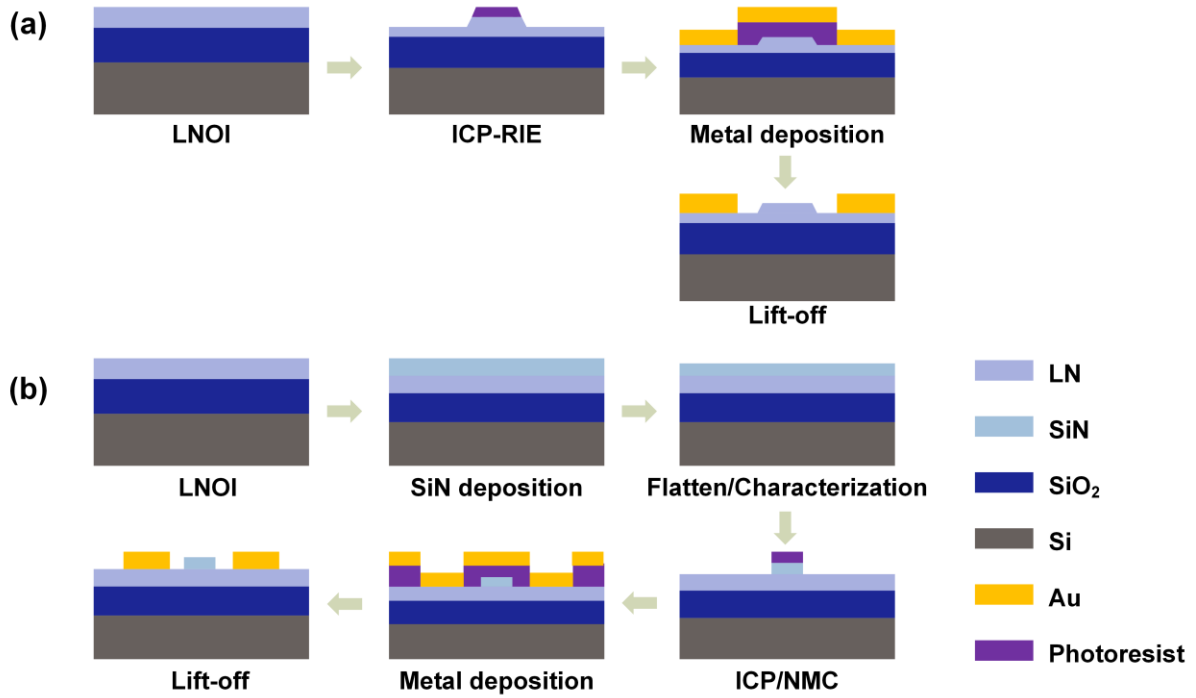


Figure S3. A comparison of the process flows. (a) Process flow of the directly etched devices on LNOI. (b) Process flow of the heterogeneously integrated devices based on SiN on LNOI.

4. Comparison of optical mode distributions in directly etched and heterogeneously integrated waveguide structures

The TPhC waveguide structure is composed of two regions, slab A and slab B. To investigate the optical confinement characteristics, we simulated the optical mode distributions for both directly etched and heterogeneously integrated waveguides, as illustrated in Figure S4. In the directly etched configuration, the optical mode is well confined within the LN layer, whereas in the heterogeneous integration scheme, a portion of the mode extends into the SiN cladding. This partial leakage into the SiN layer reduces the modal overlap with the LN core. Given that the modulation mechanism is governed by the linear electro-optic (EO) effect of LN, the directly etched TPhC waveguide offers superior EO performance due to its enhanced optical–electrical field interaction within the active material.

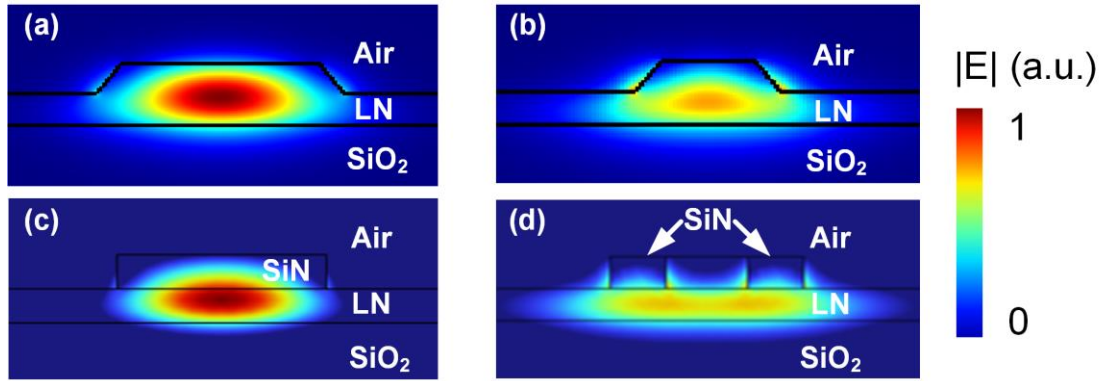


Figure S4. A comparison of the mode profiles. Mode distributions in (a) slabs A and (b) B of the directly etched TPhC waveguide, and in (c) slabs A and (d) B of the heterogeneously integrated waveguide structures.

5. The slow light effect in the TPhC structure

The TPhC cavity plays a central role in enhancing light confinement. However, in our device, the slow-light effect inherent to the TPhC structure also contributes to enhanced light–electric field interaction. Specifically, although our structure operates as a cavity, the slow light effect within the TPhC cavity leads to a reduced group velocity at the resonance, as shown in Figure S5. The increase in group delay induced by the TPhC microcavities leads to a reduction in the group velocity, thereby resulting in the slow-light effect. This results in a significantly increased group index, effectively prolonging the interaction time between the optical and electrical fields within the same physical length. In our case, both effects are present: the TPhC cavity ensures strong optical field localization, while the slow-light effect further enhances the light–electric field interaction through temporal stretching of the optical mode.

We conducted simulations of the TPhC structure with varying numbers of unit cells (N), ranging from $N = 50$ to $N = 90$. For each configuration, the group delay was calculated to extract the group velocity, from which the corresponding group index n_g was determined. The simulation results reveal that the proposed structure exhibits a pronounced slow-light effect, significantly enhancing light–matter interaction. Moreover, as the number of unit cells increases, the optical confinement at the resonant wavelength becomes stronger, leading to a further increase in the group index n_g .

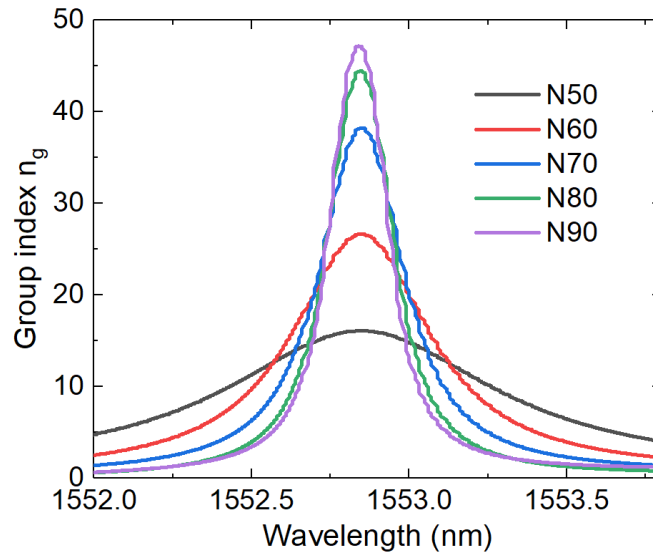


Figure S5. Slow light effect in the TPhC structure. The simulated group index n_g of the TPhC structure when unit cell number from 50 to 90.